Company-Based Python Problems (List, Tuple, Dictionary, Set, Arrays)

1. Employee Salary Analysis (List)

```
Problem: Calculate average and highest salary from a list.

Input:
5
25000 30000 28000 35000 40000

Output:
Average Salary: 31600
Highest Salary: 40000

Code:
n = int(input())
salaries = list(map(int, input().split()))
avg = sum(salaries) / n
highest = max(salaries)
print("Average Salary:", int(avg))
print("Highest Salary:", highest)
```

2. Product Inventory System (Dictionary)

```
Problem: Store product prices and calculate total inventory value.

Input:
3
Laptop 50000
Mouse 500
Keyboard 1500

Output:
Total Inventory Value: 52000

Code:
n = int(input())
inventory = {}
for _ in range(n):
    name, price = input().split()
    inventory[name] = int(price)
total = sum(inventory.values())
print("Total Inventory Value:", total)
```

3. Unique Website Visitors (Set)

```
Problem: Count unique visitor IDs.

Input:
7
101 102 103 101 104 102 105

Output:
Unique Visitors: 5

Code:
n = int(input())
visitors = list(map(int, input().split()))
unique = set(visitors)
print("Unique Visitors:", len(unique))
```

4. Student Record System (Tuple)

```
Problem: Print students who scored above 80.

Input:
3
101 Rahul 78
102 Anjali 92
103 Ravi 85

Output:
102 Anjali
103 Ravi

Code:
n = int(input())
students = []
for _ in range(n):
    sid, name, marks = input().split()
    students.append((int(sid), name, int(marks)))
for s in students:
    if s[2] > 80:
        print(s[0], s[1])
```

5. Sales Data Analysis (Array/List)

```
Problem: Find the day with maximum sales.

Input:
5
1200 1500 900 2000 1800

Output:
Max Sales: 2000
Day: 4

Code:
n = int(input())
sales = list(map(int, input().split()))
max_sales = max(sales)
day = sales.index(max_sales) + 1
print("Max Sales:", max_sales)
print("Day:", day)
```

6. Duplicate Order Detection (Set)

```
Problem: Count duplicate order IDs.

Input:
6
201 202 203 201 204 202

Output:
Duplicate Orders: 2

Code:
n = int(input())
orders = list(map(int, input().split()))
duplicates = n - len(set(orders))
print("Duplicate Orders:", duplicates)
```

7. Frequency of Words (Dictionary)

```
Problem: Count frequency of each word.

Input:
hello world hello python world

Output:
hello : 2
world : 2
python : 1

Code:
words = input().split()
freq = {}
for w in words:
    freq[w] = freq.get(w, 0) + 1
for k, v in freq.items():
    print(k, ":", v)
```

8. Student Marks Sorting (List)

```
Problem: Sort students by marks in descending order.

Input:
3
Rahul 78
Anjali 92
Ravi 85

Output:
Anjali 92
Ravi 85
Rahul 78

Code:
n = int(input())
students = []
for _ in range(n):
    name, marks = input().split()
    students.append((name, int(marks)))
students.sort(key=lambda x: x[1], reverse=True)
for s in students:
    print(s[0], s[1])
```

9. Common Elements in Two Arrays (Set)

```
Problem: Find common elements between two arrays.

Input:
Array1: 1 2 3 4 5
Array2: 3 4 5 6 7

Output:
Common Elements: {3, 4, 5}

Code:
a = set(map(int, input().split()))
b = set(map(int, input().split()))
common = a.intersection(b)
print("Common Elements:", common)
```